

Math 340 Midterm 1 Information and Review

- Midterm 1 date, time, and room information are posted on Canvas. It will cover Chapters 1-10 in your textbook.
- Midterm 1 is a 90-minute exam consisting of True/False questions, multiple choice problems, and free response questions.
- The only permitted materials are: pencil/pen. Any other materials/resources are prohibited.
- This review is intended to give you extra practice with connecting the concepts covered in these sections. The problems on the exam are not guaranteed to be similar to these examples.
- For additional practice, we recommend going through your homework problems, book problems/examples, discussion worksheets, and lecture examples.

Terms to know: Linear Equation, System of Linear Equations, Solution, Solution set, Consistent/Inconsistent system, parametric equation, matrix, augmented matrix, matrix entry, row, column, elementary row operations, row echelon form (REF), reduced row echelon form (RREF), leading one/pivot, row reduction, row-equivalent matrices, rank, leading variables, free variables, general solution, homogeneous system, coefficient matrix, trivial/nontrivial solution, pivot, (i,j) th entry of a matrix, size/dimension of a matrix, equal matrices, matrix addition, scalar multiple, linear combination, zero matrix, row vector, column vector, vector of unknowns, vector of constants, vector form, matrix multiplication, square matrix, main diagonal, transpose, identity matrix, inverse, invertible matrix, singular matrix, elementary matrix, scalar matrix, diagonal matrix, upper triangular matrix, lower triangular matrix, symmetric matrix, skew-symmetric matrix, minor, cofactor, cofactor expansion along a row/column, determinant

Learning Goals for Midterm 1

Clicking on each section will refer you to corresponding page in the review packet.

- Be able to set up an augmented matrix for a linear system of equations AND be able to interpret an augmented matrix as a linear system of equations
- Be able to perform elementary row operations to a matrix.
- Know how to find a REF or the RREF form for a matrix
- Be able to solve linear system using the method of Gaussian Elimination.
- Be able to find the rank of a matrix and know what it tells you about the number of solutions for the corresponding linear system.
- Be able to pick out an element of a matrix given its indices
- Know what it means for two matrices to be equal vs. row equivalent
- Know how to perform matrix operations: sum/difference, scalar multiplication, transpose, matrix-vector product (both as a linear combination and using dot product), matrix multiplication (both as repeated matrix-vector multiplication and using dot product)
- Know which sizes of matrices can be multiplied together, and what the resulting product matrix's size will be.
- Be familiar with the algebraic properties of matrix operations.
- Be familiar with properties of inverses of matrices: uniqueness, inverse of a product, inverse of inverse, inverse of transpose.
- Know how to use matrix inverses to solve linear systems of equations
- Be able to compute inverse matrices directly using row reduction.
- Know the 2×2 inverse formula
- Understand how to move between a row reduction and an elementary matrix
- Understand how to decompose a matrix into elementary matrices
- Be familiar with the special forms of square matrices (diagonal, scalar, upper/lower triangular, symmetric, and skew-symmetric) and their algebraic properties
- Know the equivalent properties to determine if a matrix is invertible (see "Box of Facts")
- Be able to directly compute determinants of small (2×2 and 3×3 , and 4×4) matrices using cofactor expansion
- Be familiar with the properties of determinants
- Be able to compute determinants of larger matrices using properties of determinants

Explanations and Examples

We can expand on the important concepts mentioned in the review packet as follows.

- Be able to set up an augmented matrix for a linear system of equations AND be able to interpret an augmented matrix as a linear system of equations

Tip: For example, suppose we have a system of linear equations in 4 variables

$$\begin{cases} 3x + 4y + 5z - 2w &= 5 \\ 3x - 2z &= 23 \\ 4y + 3z + w &= -3 \end{cases}$$

Using matrix-vector multiplication, the system of linear equations can be represented as

$$\begin{bmatrix} 3 & 4 & 5 & -2 \\ 3 & 0 & -2 & 0 \\ 0 & 4 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ 23 \\ -3 \end{bmatrix}$$

Using the augmented matrix, the system of linear equations can be represented as

$$\left[\begin{array}{cccc|c} 3 & 4 & 5 & -2 & 5 \\ 3 & 0 & -2 & 0 & 23 \\ 0 & 4 & 3 & 1 & -3 \end{array} \right]$$

- Be able to perform elementary row operations to a matrix.

Tip: There are 3 types of elementary row operations.

- Type 1: Interchange / Swap any two rows ($r_i \leftrightarrow r_j$).
- Type 2: Multiply a row by a nonzero constant ($cr_i \rightarrow r_i$).
- Type 3: Add a multiple of one row to another ($r_i + cr_j \rightarrow r_i$).

It is important to clearly indicate which row operation you are implementing.

Warning: Careful about your row operations. $2r_1 + r_2 \rightarrow r_1$ is NOT an elementary row operation. It is the combination of two elementary row operations: $2r_1 \rightarrow r_1$, THEN $r_1 + r_2 \rightarrow r_1$. This may not seem important when solving linear systems, but this distinction is important in other situations (ex: the determinant).

- Know how to find a REF or the RREF form for a matrix

Tip: An $m \times n$ matrix A is in row echelon form (REF) if it satisfies:

- All zero rows, if any, appear at the bottom of the matrix.
- The first nonzero entry from the left of a nonzero row is 1, called the leading one of its row.
- For each nonzero row, the leading one appears to the right and below any leading ones in preceding rows.

An $m \times n$ matrix A is in reduced row echelon form (RREF) if it satisfies

- A is in REF.
- If a column contains a leading one, then all the other entries in that column are zero.

For example, the matrix A is in REF but not in RREF, whereas the matrix B is in RREF.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Be able to solve linear system using the method of Gaussian Elimination.

Tip: Using augmented matrices, one can solve a linear system $A\mathbf{x} = \mathbf{b}$ as follows.

1. Convert the linear system to the augmented matrix $[A|\mathbf{b}]$.
2. Use elementary row operations to obtain a matrix $[C|\mathbf{d}]$ in REF.
3. Solve $C\mathbf{x} = \mathbf{d}$ using back substitution. You can keep reducing to RREF to get to a simpler system. The same solution solves $A\mathbf{x} = \mathbf{b}$.

- Be able to find the rank of a matrix and know what it tells you about the number of solutions for the corresponding linear system.

Tip: The **rank** of a matrix A is the number of leading ones (i.e. number of pivots) in ANY row-echelon matrix that is row-equivalent to A . Note that the rank of any matrix is bounded above by the number of rows and the number of columns of A .

For example, $M = \begin{bmatrix} -1 & 3 \\ 2 & 4 \\ 0 & -7 \end{bmatrix}$, I can do elementary row operations to get to a row equivalent matrix, $N = \begin{bmatrix} 1 & 4 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Since N is a REF matrix, I can say that the rank of M (and N) is 2.

As another example, if $C = \begin{bmatrix} 1 & -1 & 3 \\ 2 & -2 & 6 \\ -3 & 3 & -9 \end{bmatrix}$, and I do elementary row operations to reduce C to

$\begin{bmatrix} 1 & -1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (which is in RREF), then I can say that the rank of C is just 1.

Tip: For a given **consistent** linear system in m equations and n unknowns:

- If the rank of the augmented matrix is n , then every variable corresponds to a leading one. Thus, the system has a unique solution.
- If the rank = r of the augmented matrix is less than n , then there are $n - r$ free variables, so the system has infinitely many solutions.

Note: For a consistent system, the rank can't be more than n (why?).

Tip: Since every homogeneous system is consistent, if a homogeneous system has free variables, it has infinitely many solutions.

For example, if my homogeneous system is given by $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & 1 & -1 & 5 \\ 0 & 1 & 2 & 4 \\ 1 & 3 & 3 & 13 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix},$$

Then I row reduce A (technically, the augmented matrix $[A : \mathbf{0}]$, but since the zero column doesn't change, it doesn't affect the final answer) to get

$$\begin{bmatrix} 1 & 0 & -3 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This means: x_1, x_2 are leading variables, while x_3, x_4 are free. I set the free variables equal to parameters $x_3 = s, x_4 = t$. The row $[1 \ 0 \ -3 \ 1]$ tells us $x_1 - 3x_3 + x_4 = 0$, so $x_1 = 3x_3 - x_4 = 3s - t$. The row $[0 \ 1 \ 2 \ 4]$ tells us $x_2 + 2x_3 + 4x_4 = 0$, so $x_2 = -2x_3 - 4x_4 = -2s - 4t$. My solution set looks like:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3s - t \\ -2s - 4t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$$

The solutions are built from linear combinations of $\begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$, Notice that A has rank 2, and the number of parameters = number of free variables = $4 - 2$.

- Be able to pick out an element of a matrix given its indices

Tip: The (i, j) -th entry of an $m \times n$ matrix A is the entry a_{ij} on the i -th row and j -th column.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{n2} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

- Know what it means for two matrices to be equal vs. row equivalent

Tip: Two $m \times n$ matrices A and B are

- equal: if every (i, j) -th entry of A is equal to the corresponding (i, j) -th entry of B .
- row equivalent: if B can be produced by applying a finite sequence of elementary row operations to A .

Note that every $m \times n$ matrix A is row equivalent to a **unique** matrix in RREF.

- Know how to perform matrix operations: sum/difference, scalar multiplication, transpose, matrix-vector product (both as a linear combination and using dot product), matrix multiplication (both as repeated matrix-vector multiplication and using dot product)

Tip: Given two $m \times n$ matrices A and B ,

- $A + B$: Add (i, j) -th entries of A and B together

- $A - B$: Subtract (i, j) -th entries of A by (i, j) -th entries of B .
- cA : Multiply all the entries of A by a constant c .

Tip: Given an $m \times n$ matrix A , we can perform the matrix-vector multiplication $A\mathbf{x}$ (where $\mathbf{x}$ is an $n \times 1$ matrix/vector) to form an $m \times 1$ matrix/vector. We can view this computation two ways:

As a linear combination: $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$, where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, and $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are

the columns of A .

As a dot product: The i th entry in the vector $A\mathbf{x}$ is computed by taking the i th row of A dotted with $\mathbf{x}$.

For example, if we want to know if $\begin{bmatrix} 4 \\ 12 \\ -14 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 4 \\ -7 \end{bmatrix}$, this means

we seek to find x_1, x_2 with

$x_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 4 \\ -7 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ -14 \end{bmatrix}$. This is the same as asking if there's a solution to the linear system written in matrix equation form as

$$\begin{bmatrix} -1 & 3 \\ 2 & 4 \\ 0 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ -14 \end{bmatrix}$$

which is ALSO the same as asking if there's a solution to the linear system

$$-1x_1 + 3x_2 = 4 \tag{2}$$

$$2x_1 + 4x_2 = 12 \tag{3}$$

$$-7x_2 = -14 \tag{4}$$

We can solve this by setting up the augmented matrix:

$$\left[\begin{array}{cc|c} -1 & 3 & 4 \\ 2 & 4 & 12 \\ 0 & -7 & -14 \end{array} \right]$$

and reducing to:

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right]$$

This means that $x_1 = 2, x_2 = 2$ is a solution, so the answer is yes!

- Know which sizes of matrices can be multiplied together, and what the resulting product matrix's size will be.

Tip: The following matrices can be multiplied together for any integer n, m, k .

- A : An $n \times m$ matrix.

- B : An $m \times k$ matrix.
- AB : Gives an $n \times k$ matrix. The (i, j) -th entry of AB is the dot product of the i -th row of A with the j -th column of B . We can also view this as repeated matrix-vector multiplication: $AB = [A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_k]$, where $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$ are the columns of B .
- BA : Can be multiplied if $n = k$. Gives an $m \times m$ matrix. (If $n \neq k$, then BA is not well defined).

Here is an example of a multiplication of two matrices with compatible sizes.

$$\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 & -2 \\ 2 & -1 & 6 \end{bmatrix} = \begin{bmatrix} 0 \cdot 3 + 1 \cdot 2 & 0 \cdot 5 + 1 \cdot -1 & 0 \cdot -2 + 1 \cdot 6 \\ 2 \cdot 3 + 1 \cdot 2 & 2 \cdot 5 + 1 \cdot -1 & 2 \cdot -2 + 1 \cdot 6 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 8 & 11 & 2 \end{bmatrix} \quad (5)$$

- Be familiar with the algebraic properties of matrix operations.

Tip: Recall the Transpose of a matrix:

1. Transpose (notated as A^T): a matrix obtained by interchanging the rows of A to columns and the columns of A to rows.

- The transpose of an $n \times m$ matrix is an $m \times n$ matrix.

– For example, if $A = \begin{bmatrix} -1 & 2 & 0 \\ 3 & 4 & -7 \end{bmatrix}$, $A^T = \begin{bmatrix} -1 & 3 \\ 2 & 4 \\ 0 & -7 \end{bmatrix}$

Tip: Recall the following algebraic relations of **matrix addition**.

1. Commutative: $A + B = B + A$
2. Associative: $(A + B) + C = A + (B + C)$
3. Zero matrix: $A + O = A$. (i.e. the matrix whose entries are all zero)
4. Negative: $A + (-A) = O$.

Tip: Recall the following algebraic relations of **scalar multiplication**.

1. $r(sA) = rsA$
2. $(r + s)A = rA + sA$
3. $r(A + B) = rA + rB$
4. $A(rB) = r(AB)$

Tip: Recall the following algebraic relations of **matrix multiplication**.

1. Associative: $A(BC) = (AB)C$
2. Distributive: $A(B + C) = AB + AC$, $(A + B)C = AC + BC$.
3. Identity matrix I_n : $AI_n = A$. (i.e. the $n \times n$ square matrix whose diagonal entries are equal to 1 and all other entries are zero.)
4. **Inverse matrix**: A unique matrix A^{-1} associated to an **invertible square** $n \times n$ matrix A such that $AA^{-1} = I_n$.
5. **NOT Commutative**: $AB \neq BA$.

Tip: Recall the following algebraic relations of **transpose**.

1. $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$.
3. $(AB)^T = B^T A^T$ (**Order of multiplication changes**).
4. $(rA)^T = rA^T$.

Tip: Recall Matrix Power: A^n , multiply a square matrix A with itself by n times.

(**Warning:** $(AB)^n \neq A^n B^n$).

- Be familiar with properties of inverses of matrices: uniqueness, inverse of a product, inverse of inverse, inverse of transpose.

Tip: Recall the following properties of inverses.

1. Uniqueness: Every invertible matrix A has a unique inverse A^{-1} .
2. $(A^{-1})^{-1} = A$.
3. $(A^T)^{-1} = (A^{-1})^T$.
4. $(AB)^{-1} = B^{-1}A^{-1}$. (**Order of multiplication changes**)

- Know how to use matrix inverses to solve linear systems of equations

Tip: Given a square $n \times n$ matrix A , the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if A is invertible. In particular, we have $\mathbf{x} = A^{-1}\mathbf{b}$.

- Be able to compute inverse matrices directly using row reduction.

Tip: Using augmented matrices, one can find an inverse of an $n \times n$ matrix A as follows.

1. Set up the augmented matrix $[A|I]$, where I is the $n \times n$ identity matrix.
2. Use elementary row operations to obtain a matrix $[I|C]$.
3. The matrix C is the inverse of A , i.e. $C = A^{-1}$.

- Know the 2×2 inverse formula

Tip: The 2×2 inverse is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Most people remember the $\frac{1}{ad - bc}$ part, but sometimes are unsure about the matrix. Note that

$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)I_2$, so if we multiply both sides by $\frac{1}{ad - bc}$, we get $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \left(\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right) = I_2$

- Understand how to move between a row reduction and an elementary matrix

Tip: Recall the following correspondence between elementary row operation and matrix multiplication.

{Elementary row operation on A } $\iff$ {Left multiplication by elementary matrix E , i.e. EA }

The elementary matrix E can be obtained as follows.

{Elementary matrix E of type 1,2,3 } $\iff$ {Apply type 1,2,3 operation to the identity matrix I }

For example, an elementary row operation corresponding to multiplying row 3 of a 3×3 matrix A by 4 can be represented as an elementary matrix $E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$, obtained from multiplying the row 3 of the identity matrix by 4.

- Understand how to decompose a matrix into elementary matrices

Tip: Consecutive elementary row operation on A corresponds to consecutive left multiplication on A by corresponding elementary matrices E .

For example, suppose we have two matrices $A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -1 & 14 \\ 0 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$. The matrix B can be obtained from A using three consecutive elementary row operations.

1. Operation 1: Swap row 1 and row 3 of matrix A : $\begin{pmatrix} 2 & 1 & 4 \\ 0 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$
2. Operation 2: Multiply row 1 of the matrix obtained from Operation 1 by 2: $\begin{pmatrix} 4 & 2 & 8 \\ 0 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$
3. Operation 3: Using the matrix obtained from Operation 2, add row 1 with row 2 multiplied by 3: $\begin{pmatrix} 4 & -1 & 14 \\ 0 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$

Each of the operation mentioned above corresponds to the following elementary matrix

$$E_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

One can check that $B = E_3 E_2 E_1 A$ (**Remember to multiply consecutively to the left of the matrix A , not to the right**).

- Be familiar with the special forms of square matrices (diagonal, scalar, upper/lower triangular, symmetric, and skew-symmetric) and their algebraic properties

Tip: Recall the following types of matrices.

1. Symmetric: A square matrix such A that $A = A^T$.
2. Skew-Symmetric: A square matrix A such that $A^T = -A$.
3. Scalar matrix: An $n \times n$ square matrix A such that $A = cI_n$ for some scalar c .
4. Diagonal: A square matrix such that all entries other than the main diagonal entries are 0.
5. Upper triangular: A square matrix such that $a_{ij} = 0$ for $i > j$.
6. Lower triangular: A square matrix such that $a_{ij} = 0$ for $i < j$.

See <https://sites.ualberta.ca/~jsylvest/books/DLA/section-special-forms-theory.html> for a summary of the algebraic properties of these special forms.

For example, we can show that if A, B are any 2×2 symmetric matrices, then their sum $A + B$ must be symmetric, but their product AB may not be symmetric.

To show $A + B$ is symmetric: If A and B are symmetric, then $A^T = A$ and $B^T = B$. We want to show that $C = A + B$ is also symmetric, so we take the transpose: $C^T = (A + B)^T = A^T + B^T$. Since $A^T = A$ and $B^T = B$, this means $(A + B)^T = A^T + B^T = A + B = C$. So $A + B$ is also symmetric.

To demonstrate AB may not be symmetric, we find explicit matrices A and B which are symmetric but their product AB is not. There are many options. I use $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$. A and B are both symmetric, but $AB = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$ is not, since $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$.

- Know the equivalent properties to determine if a matrix is invertible (i.e. “Box of Facts”)

Tip: The following statements are equivalent for a square $n \times n$ matrix A (i.e. this is our current “Box of Facts”): Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n .
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be expressed as a product of some number of elementary matrices.
7. $\det(A) \neq 0$.

- Be able to directly compute determinants of small (2×2 and 3×3 , and 4×4) matrices using cofactor expansion

Tip: The determinant of an $n \times n$ matrix A can be obtained from

$$\det(A) = \sum_{j=1}^n a_{ij} \text{cof}(A)_{ij} = \sum_{j=1}^n a_{ij} (-1)^{i+j} \text{minor}(A)_{ij}$$

where the (i, j) -th minor $\text{minor}(A)_{ij}$ is the determinant of the $(n - 1) \times (n - 1)$ matrix which results from deleting the i -th row and the j -th column of A .

Remember that you can choose any row/column for the cofactor expansion, so choose one that makes your computation easy.

For example, if we want to compute the determinant of $A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -3 & 2 & 2 \\ 0 & 5 & 0 & 0 \\ 7 & 3 & 2 & -1 \end{bmatrix}$, we should choose

row 3 for the cofactor expansion. Since 5 is the only nonzero entry, and it's in the $(3, 2)$ entry, so the sign will be $-1 = (-1)^{3+2}$, our computation becomes

$\det \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -3 & 2 & 2 \\ 0 & 5 & 0 & 0 \\ 7 & 3 & 2 & -1 \end{bmatrix} = (-5) \det \begin{bmatrix} 1 & 0 & 3 \\ 1 & 2 & 2 \\ 7 & 2 & -1 \end{bmatrix}$. I'll then use row 1 for my cofactor expansion of the 3×3 matrix:

$$(-5) \det \begin{bmatrix} 1 & 0 & 3 \\ 1 & 2 & 2 \\ 7 & 2 & -1 \end{bmatrix} = -5 \left((1) \det \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix} + (3) \det \begin{bmatrix} 1 & 2 \\ 7 & 2 \end{bmatrix} \right) = -5(-6 - 36) = 210.$$

The determinant of A is 210.

Another example: Consider the matrix

$$A = \begin{bmatrix} 3 & 4 & 5 & 2 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 3 & 0 \\ 5 & 2 & 8 & 9 \end{bmatrix}$$

We note that the second row of the matrix has (2,3)-th entry as 1 whereas all the other entries are 0. This allows us to simplify the determinant as

$$\det(A) = a_{23} \cdot (-1)^{2+3} \text{minor}(A)_{23} = -\det \begin{pmatrix} 3 & 4 & 2 \\ -1 & 0 & 0 \\ 5 & 2 & 9 \end{pmatrix}$$

Again, the second row of the matrix $\begin{bmatrix} 3 & 4 & 2 \\ -1 & 0 & 0 \\ 5 & 2 & 9 \end{bmatrix}$ has (2,1)-th entry as -1 whereas all the other entries are 0. This allows us to simplify the determinant as

$$\det \begin{bmatrix} 3 & 4 & 2 \\ -1 & 0 & 0 \\ 5 & 2 & 9 \end{bmatrix} = a_{21} \cdot (-1)^{2+1} \det \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix} = (-1) \cdot (-1) \cdot (4 \cdot 9 - 2 \cdot 2) = 32.$$

Therefore, we have

$$\det(A) = -\det \begin{bmatrix} 3 & 4 & 2 \\ -1 & 0 & 0 \\ 5 & 2 & 9 \end{bmatrix} = -32.$$

- Be familiar with the properties of determinants

Tip: Recall the following properties of determinants of elementary row operations.

1. Type 1: $\det(A_{r_i \leftrightarrow r_j}) = -1 \cdot \det(A)$.
Interchanging / swapping row r_i and r_j multiplies the determinant by -1 .
2. Type 2: $\det(A_{kr_i \rightarrow r_i}) = k \det(A)$
Multiplying row r_i by k also multiplies the determinant by k .
3. Type 3: $\det(A_{kr_i + r_j \rightarrow r_j}) = \det(A)$
Adding to row r_j k times the row r_i preserves the determinant.

Tip: Recall the following properties of determinants of arbitrary $n \times n$ matrices.

1. If A is an upper / lower triangular $n \times n$ matrices, then $\det(A) = a_{11}a_{22} \cdots a_{nn}$, i.e. the determinant of a triangular matrix is the product of the elements on the main diagonal.
2. We have $\det(A) = \det(A^T)$.
3. If a row of A consists entirely of zeros, then $\det(A) = 0$.

4. For any scalar k and an $n \times n$ matrix A , we have $\det(kA) = k^n \det(A)$.
5. If A, B are two $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.
6. If A is non-singular, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

- Be able to compute determinants of larger matrices using properties of determinants

Tip: For example, suppose A and B are invertible matrices such that $\det(A) = 3$ and $\det(B) = 2$. Let's try to compute the determinant of the matrix $A^3 B^T A^{-2}$. We have

$$\det(A^3 B^T A^{-2}) = \det(A^3) \det(B^T) \det(A^{-2}) = \det(A)^3 \det(B) \det(A)^{-2} = \det(A) \det(B) = 6.$$

Tip: For example, suppose the determinant of the following matrix A is equal to 3.

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = 3.$$

Let's try to compute the determinant of the following matrix B :

$$\det \begin{bmatrix} a_1 + 2b_1 - 3c_1 & a_2 + 2b_1 - 3c_1 & a_3 + 2b_3 - 3c_3 \\ c_1 & c_2 & c_3 \\ -2b_1 & -2b_2 & -2b_3 \end{bmatrix}$$

Note that the following consecutive elementary row operations on A gives the matrix B .

1. Operation 1: Swap row 2 and row 3 of the matrix A . This creates the new matrix B_1 , where $\det(B_1) = -3$
2. Operation 2: Multiply row 3 of the matrix from Operation 1 by -2 . This creates the new matrix B_2 , where $\det(B_2) = (-3)(-2) = 6$.
3. Operation 3: Subtract row 1 by row 3 of the matrix from Operation 2 (replacing row 3). This creates the new matrix B_3 , with same determinant as B_2 . $\det(B_3) = 6$.
4. Operation 4: Add row 1 by 3 times the row 2 of the matrix from Operation 3. This creates the new matrix B_4 , with same determinant as B_3 . $\det(B_4) = 6$.

Because $B = B_4$, we have $\det(B) = 6$.

Tip: We show that if n is odd, then any skew symmetric $n \times n$ matrix A satisfies $\det(A) = 0$. Recall that A is skew symmetric if $A = -A^T$. Then we have

$$\det(A) = \det(-1 \cdot A^T) = (-1)^n \det(A^T) = (-1)^n \det(A)$$

Because n is odd, we have

$$\det(A) = (-1) \cdot \det(A)$$

Therefore, we have $\det(A) = 0$.